Deployment Process Document

With CI/CD Pipeline & UAT Environment

Project: REB AMI — Reporting & Customer Portal

Infrastructure: 6 VMs Production + 3 VMs UAT | On-premise Data Center

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1. Environment Overview

1.1 Environment Summary

Environment	Purpose	VMs	URL
Development	Local development	Developer machines	localhost
UAT	User Acceptance Testing	3 VMs	uat.ami.reb.gov.bd
Production	Live system	6 VMs	ami.reb.gov.bd

1.2 Environment Architecture

```
┌──────────────────────────────────────────────────────────────────┐
│                          DEVELOPERS                              │
│                (Local Development Environment)                   │
└──────────────────────────┬───────────────────────────────────────┘
                           │ Git Push
                           ▼
┌──────────────────────────────────────────────────────────────────┐
│                      GITLAB / GITHUB                             │
│                     (Source Control)                             │
└──────────────────────────┬───────────────────────────────────────┘
                           │ Webhook Trigger
                           ▼
┌──────────────────────────────────────────────────────────────────┐
│                      JENKINS SERVER                              │
│                    (CI/CD Pipeline)                              │
└──────────┬──────────────────────────────────┬────────────────────┘
           │                                  │
```

```
                ▼                                   ▼

┌──────────────────────┐              ┌──────────────────────┐
│   UAT ENVIRONMENT    │              │ PRODUCTION ENV       │
│     (3 VMs)          │              │    (6 VMs)           │
│  Auto Deploy         │──────────────▶│  Manual Approval     │
└──────────────────────┘   Promote    └──────────────────────┘
```

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2. UAT Environment Setup (3 VMs)

2.1 UAT VM Specifications

VM	Role	vCPU	RAM	Storage	IP Address
UAT-VM-01	App + Web + LB	4	8 GB	100 GB	10.0.10.11
UAT-VM-02	PostgreSQL	4	8 GB	100 GB	10.0.10.12
UAT-VM-03	Jenkins CI/CD	4	8 GB	100 GB	10.0.10.13
TOTAL		**12**	**24 GB**	**300 GB**	

2.2 UAT Architecture

```
┌──────────────────────────────────────────────────────────────┐
│                      UAT ENVIRONMENT                         │
│                                                              │
│   ┌──────────────────────────────────────────────────────┐   │
│   │              UAT-VM-01 (All-in-One)                  │   │
│   │   ┌──────────┐  ┌──────────┐  ┌──────────────────┐  │   │
│   │   │  NGINX   │  │ Node.js  │  │  Vue.js Dashboard  │  │   │
│   │   │   LB     │  │   API    │  │   (Static Files)   │  │   │
│   │   └──────────┘  └──────────┘  └──────────────────┘  │   │
│   └──────────────────────────────────────────────────────┘   │
│                            │                                 │
│                            ▼                                 │
│   ┌──────────────────────────────────────────────────────┐   │
│   │              UAT-VM-02 (Database)                    │   │
│   │                    PostgreSQL                        |   │
│   └──────────────────────────────────────────────────────┘   │
│                            │                                 │
│                            ▼                                 │
```

```
│                    Oracle DB (External)                        │
└────────────────────────────────────────────────────────────────┘
```

---

3. CI/CD Pipeline Setup

3.1 Jenkins Server Setup (UAT-VM-03)

```bash
# 1. Install Java
sudo apt update
sudo apt install openjdk-17-jdk -y

# 2. Add Jenkins repository
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee \
  /usr/share/keyrings/jenkins-keyring.asc > /dev/null

echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
  https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null

# 3. Install Jenkins
sudo apt update
sudo apt install jenkins -y

# 4. Start Jenkins
sudo systemctl start jenkins
sudo systemctl enable jenkins

# 5. Get initial admin password
sudo cat /var/lib/jenkins/secrets/initialAdminPassword

# 6. Install required plugins
# - Git
# - NodeJS
# - Pipeline
# - SSH Agent
# - Slack Notification (optional)
```

3.2 Jenkins Pipeline Configuration

Jenkinsfile (API - Node.js):

groovy

```
pipeline {
    agent any

    environment {
        NODE_VERSION = '20'
        UAT_SERVER = '10.0.10.11'
        PROD_APP1 = '10.0.2.11'
        PROD_APP2 = '10.0.2.12'
        PROD_WEB = '10.0.2.13'
    }

    tools {
        nodejs "${NODE_VERSION}"
    }

    stages {

        // ========== BUILD STAGE ==========
        stage('Checkout') {
            steps {
                checkout scm
                echo "Branch: ${env.BRANCH_NAME}"
            }
        }

        stage('Install Dependencies') {
            steps {
                sh 'npm ci'
            }
        }

        stage('Lint') {
            steps {
                sh 'npm run lint'
            }
        }

        stage('Unit Tests') {
            steps {
                sh 'npm run test:unit'
            }
            post {
                always {
```

```
                junit 'test-results/*.xml'
            }
        }
    }

    stage('Build') {
        steps {
            sh 'npm run build'
        }
    }

    // ========== UAT DEPLOYMENT ==========
    stage('Deploy to UAT') {
        when {
            branch 'develop'
        }
        steps {
            echo 'Deploying to UAT environment...'
            sshagent(['uat-server-key']) {
                sh '''
                    ssh -o StrictHostKeyChecking=no deployer@${UAT_SERVER} << 'E
                    cd /var/www/reb-ami-api
                    git pull origin develop
                    npm ci --production
                    pm2 restart reb-ami-api
                    echo "UAT Deployment Complete!"
EOF
                '''
            }
        }
        post {
            success {
                echo 'UAT Deployment Successful!'
                // slackSend channel: '#deployments', message: "UAT Deployed: ${
            }
            failure {
                echo 'UAT Deployment Failed!'
            }
        }
    }

    // ========== UAT TESTING ==========
    stage('UAT Health Check') {
        when {
```

```
        branch 'develop'
    }
    steps {
        sh '''
            sleep 10
            curl -f http://${UAT_SERVER}:3000/api/health || exit 1
            echo "UAT Health Check Passed!"
        '''
    }
}

stage('UAT Integration Tests') {
    when {
        branch 'develop'
    }
    steps {
        sh '''
            npm run test:integration -- --env=uat
        '''
    }
}

// ========== PRODUCTION DEPLOYMENT ==========
stage('Approval for Production') {
    when {
        branch 'main'
    }
    steps {
        input message: 'Deploy to Production?',
              ok: 'Deploy',
              submitter: 'admin,devops'
    }
}

stage('Deploy to Production') {
    when {
        branch 'main'
    }
    steps {
        echo 'Deploying to Production...'

        // Deploy to App Server 1
        sshagent(['prod-server-key']) {
            sh '''
```

```
                        ssh -o StrictHostKeyChecking=no deployer@${PROD_APP1} << 'EO
                        cd /var/www/reb-ami-api
                        git pull origin main
                        npm ci --production
                        pm2 restart reb-ami-api
EOF
                    '''
                }

                // Deploy to App Server 2
                sshagent(['prod-server-key']) {
                    sh '''
                        ssh -o StrictHostKeyChecking=no deployer@${PROD_APP2} << 'EO
                        cd /var/www/reb-ami-api
                        git pull origin main
                        npm ci --production
                        pm2 restart reb-ami-api
EOF
                    '''
                }
            }
            post {
                success {
                    echo 'Production Deployment Successful!'
                }
            }
        }

        stage('Production Health Check') {
            when {
                branch 'main'
            }
            steps {
                sh '''
                    sleep 10
                    curl -f https://ami.reb.gov.bd/api/health || exit 1
                    echo "Production Health Check Passed!"
                '''
            }
        }
    }

    post {
        always {
```

```
            cleanWs()
        }
        success {
            echo 'Pipeline completed successfully!'
        }
        failure {
            echo 'Pipeline failed!'
            // slackSend channel: '#alerts', color: 'danger', message: "Build Failed
        }
    }
}
```

Jenkinsfile (Web Dashboard - Vue.js):

```groovy
```

```
pipeline {
    agent any

    environment {
        NODE_VERSION = '20'
        UAT_SERVER = '10.0.10.11'
        PROD_WEB = '10.0.2.13'
    }

    tools {
        nodejs "${NODE_VERSION}"
    }

    stages {

        stage('Checkout') {
            steps {
                checkout scm
            }
        }

        stage('Install Dependencies') {
            steps {
                sh 'npm ci'
            }
        }

        stage('Lint') {
            steps {
                sh 'npm run lint'
            }
        }

        stage('Unit Tests') {
            steps {
                sh 'npm run test:unit'
            }
        }

        stage('Build UAT') {
            when {
                branch 'develop'
            }
```

```
            steps {
                sh 'npm run build -- --mode uat'
            }
        }

        stage('Build Production') {
            when {
                branch 'main'
            }
            steps {
                sh 'npm run build -- --mode production'
            }
        }

        stage('Deploy to UAT') {
            when {
                branch 'develop'
            }
            steps {
                sshagent(['uat-server-key']) {
                    sh '''
                        rsync -avz --delete dist/ deployer@${UAT_SERVER}:/var/www/re
                        ssh deployer@${UAT_SERVER} "sudo systemctl reload nginx"
                    '''
                }
            }
        }

        stage('Approval for Production') {
            when {
                branch 'main'
            }
            steps {
                input message: 'Deploy Web Dashboard to Production?',
                      ok: 'Deploy'
            }
        }

        stage('Deploy to Production') {
            when {
                branch 'main'
            }
            steps {
                sshagent(['prod-server-key']) {
```

```
                    sh '''
                        rsync -avz --delete dist/ deployer@${PROD_WEB}:/var/www/reb-
                        ssh deployer@${PROD_WEB} "sudo systemctl reload nginx"
                    '''
                }
            }
        }
    }
}
```

---

4. Git Branching Strategy

4.1 Branch Structure

```
main (Production)
  │
  ├── develop (UAT)
  │     │
  │     ├── feature/user-auth
  │     ├── feature/billing-report
  │     ├── feature/mobile-api
  │     └── bugfix/login-issue
  │
  └── hotfix/critical-fix
```

4.2 Branch Rules

Branch	Purpose	Deploys To	Auto/Manual
`feature/*`	New features	-	-
`bugfix/*`	Bug fixes	-	-
`develop`	Integration branch	**UAT**	Auto
`main`	Production release	**Production**	Manual Approval
`hotfix/*`	Emergency fixes	Production	Manual

4.3 Git Workflow

```
1. Developer creates feature branch
   git checkout -b feature/new-report develop

2. Developer commits changes
   git add .
   git commit -m "feat: add new billing report"

3. Create Pull Request to develop
   PR: feature/new-report → develop

4. Code Review & Merge
   After approval, merge to develop

5. Auto Deploy to UAT
   Jenkins automatically deploys to UAT

6. UAT Testing
   QA team tests in UAT environment

7. Create Release PR
   PR: develop → main

8. Production Deployment
   Manual approval in Jenkins
   Deploy to Production
```

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5. UAT Testing Process

5.1 UAT Test Phases

```
┌──────────────────────────────────────────────────────────────┐
│                      UAT TESTING WORKFLOW                    │
└──────────────────────────────────────────────────────────────┘


  ┌──────────┐     ┌──────────┐     ┌──────────┐     ┌──────────┐
  │  Deploy  │──▶│  Smoke   │──▶│Functional│──▶│   UAT    │
  │  to UAT  │     │  Tests   │     │  Tests   │     │ Sign-off │
  └──────────┘     └──────────┘     └──────────┘     └──────────┘
       │                │                │                │
```

```
    ▼              ▼              ▼              ▼
  Automated      Automated      Manual +       Stakeholder
  (Jenkins)      (Jenkins)      Automated       Approval
```

5.2 UAT Test Cases

Test ID	Module	Test Case	Priority
UAT-001	Login	User login with valid credentials	High
UAT-002	Login	User login with invalid credentials	High
UAT-003	Dashboard	Dashboard loads with correct data	High
UAT-004	Bills	View customer bill history	High
UAT-005	Bills	Download bill PDF	Medium
UAT-006	Reports	Generate usage report	High
UAT-007	Reports	Export report to Excel	Medium
UAT-008	Mobile API	Fetch customer data	High
UAT-009	Mobile API	Check account balance	High
UAT-010	Search	Search customer by ID	Medium
UAT-011	Search	Search customer by name	Medium
UAT-012	Oracle Sync	Verify data from Oracle	High

5.3 UAT Sign-off Form

```
╔═══════════════════════════════════════════════════════════════════╗
||                    UAT SIGN-OFF FORM                            ||
╟═══════════════════════════════════════════════════════════════════╢
|| Release Version: ________________    Date: ________________   ||
||                                                                 ||
|| Testing Period: From _____________ To _____________            ||
||                                                                 ||
|| Total Test Cases: _______    Passed: _______   Failed: ______ ||
||                                                                 ||
|| Critical Issues Found: _______                                  ||
```

```
│ Critical Issues Resolved: _______                                  │
│                                                                    │
│ UAT Result:  [ ] APPROVED    [ ] REJECTED                          │
│                                                                    │
│ Comments: ____________________________________________________     │
│ ____________________________________________________________       │
│                                                                    │
│ Approved By:                                                       │
│                                                                    │
│ QA Lead: _________________ Signature: ________ Date: ______        │
│ Business Owner: __________ Signature: ________ Date: ______        │
│ Project Manager: _________ Signature: ________ Date: ______        │
└────────────────────────────────────────────────────────────────────┘
```

---

6. Deployment Pipeline Flow

6.1 Complete CI/CD Flow

```
┌──────────────────────────────────────────────────────────────────┐
│                          DEVELOPER                               │
│                      git push origin feature/xyz                 │
└──────────────────────────────┬───────────────────────────────────┘
                               │
                               ▼

┌──────────────────────────────────────────────────────────────────┐
│                     PULL REQUEST (PR)                            │
│               feature/xyz → develop                              │
│                                                                  │
│   ┌──────────────┐   ┌──────────────┐   ┌──────────────┐        │
│   │ Code Review  │   │  PR Checks   │   │   Approve    │        │
│   │  (Manual)    │   │  (Jenkins)   │   │   & Merge    │        │
│   └──────────────┘   └──────────────┘   └──────────────┘        │
└──────────────────────────────┬───────────────────────────────────┘
                               │ Merge to develop
                               ▼

┌──────────────────────────────────────────────────────────────────┐
│                 JENKINS PIPELINE (develop)                       │
│                                                                  │
│   ┌────────┐   ┌────────┐   ┌────────┐   ┌────────┐   ┌──────────┐   │
│   │Checkout│─▶│  Lint  │─▶│  Test  │─▶│ Build  │─▶│Deploy UAT│   │
│   └────────┘   └────────┘   └────────┘   └────────┘   └──────────┘   │
│
```

```
└──────────────────────────────────────────────────────────────┘
                                │
                                ▼

┌──────────────────────────────────────────────────────────────┐
│                     UAT ENVIRONMENT                          │
│                   uat.ami.reb.gov.bd                         │
│                                                              │
│   ┌──────────────┐   ┌──────────────┐   ┌──────────────┐    │
│   | Smoke Tests |   |  QA Testing |   |UAT Sign-off |      │
│   |  (Auto)     |   |  (Manual)   |   |  (Manual)   |      │
│   └──────────────┘   └──────────────┘   └──────────────┘    │
└──────────────────────────────────────────────────────────────┘
                                │ UAT Approved
                                ▼

┌──────────────────────────────────────────────────────────────┐
│                     PULL REQUEST (PR)                        │
│                  develop → main                              │
│                                                              │
│   ┌──────────────┐   ┌──────────────┐   ┌──────────────┐    │
│   | Code Review |   |Release Notes|   |   Approve   |      │
│   |  (Manual)   |   |   Created   |   |   & Merge   |      │
│   └──────────────┘   └──────────────┘   └──────────────┘    │
└──────────────────────────────────────────────────────────────┘
                                │ Merge to main
                                ▼

┌──────────────────────────────────────────────────────────────┐
│                 JENKINS PIPELINE (main)                      │
│                                                              │
│   ┌────────┐   ┌────────┐   ┌────────┐   ┌──────────────────────┐   │
│   |  Test  |─▶| Build  |─▶|Approval|─▶|  Deploy Production   |  |
│   |        |   |        |   |(Manual)|   |  (Rolling Update)    |  |
│   └────────┘   └────────┘   └────────┘   └──────────────────────┘   │
└──────────────────────────────────────────────────────────────┘
                                │
                                ▼

┌──────────────────────────────────────────────────────────────┐
│                 PRODUCTION ENVIRONMENT                       │
│                    ami.reb.gov.bd                            │
│                                                              │
│   ┌──────────────┐   ┌──────────────┐   ┌──────────────┐    │
│   |Health Check |   |  Monitor    |   |   LIVE!     |      │
│   |  (Auto)     |   |  (Ongoing)  |   |             |      │
```

```
│   └──────────────┘   └──────────────┘   └──────────────┘                │
└──────────────────────────────────────────────────────────────────────────┘
```

---

7. Production Deployment Steps

7.1 Rolling Deployment Strategy

```
Step 1: Remove APP-VM-01 from Load Balancer
        ↓
Step 2: Deploy to APP-VM-01
        ↓
Step 3: Health Check APP-VM-01
        ↓
Step 4: Add APP-VM-01 back to Load Balancer
        ↓
Step 5: Remove APP-VM-02 from Load Balancer
        ↓
Step 6: Deploy to APP-VM-02
        ↓
Step 7: Health Check APP-VM-02
        ↓
Step 8: Add APP-VM-02 back to Load Balancer
        ↓
Step 9: Deploy Web Dashboard (VM4)
        ↓
Step 10: Final Health Check
```

7.2 Rolling Deployment Script

```bash
```

```bash
#!/bin/bash
# rolling-deploy.sh

set -e

NGINX_SERVER="10.0.1.10"
APP_SERVER_1="10.0.2.11"
APP_SERVER_2="10.0.2.12"
WEB_SERVER="10.0.2.13"

echo "==========================================="
echo "  REB AMI - Rolling Deployment Script"
echo "==========================================="

# Function: Remove server from NGINX
remove_from_lb() {
    local server=$1
    echo "Removing $server from load balancer..."
    ssh deployer@$NGINX_SERVER "sudo sed -i 's/server $server:3000/server $server:30
    sleep 5
}

# Function: Add server to NGINX
add_to_lb() {
    local server=$1
    echo "Adding $server to load balancer..."
    ssh deployer@$NGINX_SERVER "sudo sed -i 's/server $server:3000 down/server $serv
    sleep 5
}

# Function: Deploy to app server
deploy_app() {
    local server=$1
    echo "Deploying to $server..."
    ssh deployer@$server << 'EOF'
        cd /var/www/reb-ami-api
        git pull origin main
        npm ci --production
        pm2 restart reb-ami-api
EOF
}

# Function: Health check
```

```
health_check() {
    local server=$1
    echo "Health check for $server..."
    for i in {1..10}; do
        if curl -sf http://$server:3000/api/health > /dev/null; then
            echo "✓ $server is healthy"
            return 0
        fi
        sleep 3
    done
    echo "✗ $server health check failed!"
    exit 1
}

# Deploy to App Server 1
echo ""
echo ">>> Deploying to App Server 1..."
remove_from_lb $APP_SERVER_1
deploy_app $APP_SERVER_1
health_check $APP_SERVER_1
add_to_lb $APP_SERVER_1

# Deploy to App Server 2
echo ""
echo ">>> Deploying to App Server 2..."
remove_from_lb $APP_SERVER_2
deploy_app $APP_SERVER_2
health_check $APP_SERVER_2
add_to_lb $APP_SERVER_2

# Deploy Web Dashboard
echo ""
echo ">>> Deploying Web Dashboard..."
ssh deployer@$WEB_SERVER << 'EOF'
    cd /var/www/reb-ami-web
    git pull origin main
    npm ci
    npm run build
    sudo systemctl reload nginx
EOF

echo ""
echo "==========================================="
```

```
echo "  Deployment Complete!"
echo "========================================="
```

---

8. Rollback Procedure

8.1 Quick Rollback Steps

```bash
```

```bash
#!/bin/bash
# rollback.sh

VERSION=$1  # e.g., v1.2.3 or commit hash

if [ -z "$VERSION" ]; then
    echo "Usage: ./rollback.sh <version>"
    exit 1
fi

echo "Rolling back to version: $VERSION"

# Rollback App Servers
for server in 10.0.2.11 10.0.2.12; do
    echo "Rolling back $server..."
    ssh deployer@$server << EOF
        cd /var/www/reb-ami-api
        git checkout $VERSION
        npm ci --production
        pm2 restart reb-ami-api
EOF
done

# Rollback Web Dashboard
ssh deployer@10.0.2.13 << EOF
    cd /var/www/reb-ami-web
    git checkout $VERSION
    npm ci
    npm run build
    sudo systemctl reload nginx
EOF

echo "Rollback complete!"
```

---

9. Environment Configuration

9.1 Environment Variables

Variable	UAT	Production
`NODE_ENV`	uat	production

Variable	UAT	Production
`API_URL`	https://uat.ami.reb.gov.bd	https://ami.reb.gov.bd
`PG_HOST`	10.0.10.12	10.0.3.21
`ORACLE_HOST`	oracle-uat.reb.gov.bd	oracle.reb.gov.bd
`LOG_LEVEL`	debug	info

9.2 Vue.js Environment Files

.env.uat:

```env
VUE_APP_API_URL=https://uat.ami.reb.gov.bd/api
VUE_APP_ENV=uat
```

.env.production:

```env
VUE_APP_API_URL=https://ami.reb.gov.bd/api
VUE_APP_ENV=production
```

---

10. Monitoring & Alerts

10.1 Deployment Notifications

Event	Channel	Recipients
UAT Deploy Success	Slack #deployments	Dev Team
UAT Deploy Failed	Slack #alerts	Dev Team
Prod Deploy Started	Slack #deployments	All
Prod Deploy Success	Slack #deployments + Email	All + Management
Prod Deploy Failed	Slack #alerts + SMS	DevOps + Lead

10.2 Health Check URLs

Environment	URL	Expected
UAT API	https://uat.ami.reb.gov.bd/api/health	{"status":"ok"}
UAT Web	https://uat.ami.reb.gov.bd	HTTP 200
Prod API	https://ami.reb.gov.bd/api/health	{"status":"ok"}
Prod Web	https://ami.reb.gov.bd	HTTP 200

---

11. Deployment Checklist

11.1 Pre-Deployment

Task	UAT	Prod
[ ] Code review completed	✓	✓
[ ] All tests passing	✓	✓
[ ] UAT testing completed	-	✓
[ ] UAT sign-off received	-	✓
[ ] Release notes prepared	-	✓
[ ] Rollback plan ready	✓	✓
[ ] Team notified	✓	✓

11.2 Post-Deployment

Task	UAT	Prod
[ ] Health checks passed	✓	✓
[ ] Smoke tests passed	✓	✓
[ ] Monitoring verified	✓	✓

Task	UAT	Prod
[ ] Stakeholders notified	-	✓
[ ] Documentation updated	-	✓

12. Approval

Role	Name	Signature	Date
DevOps Engineer			
QA Lead			
Technical Lead			
Project Manager			

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